targeted insight into why a component re-renders.

- `useDebugValue` *hook*—The `useDebugValue` hook can display custom debug information in React DevTools, making it easier to monitor state or props and understand component re-renders.
- *Third-party libraries*—Tools like why-did-you-render (https://github.com/welldone-software/why-did-you-render) offer sophisticated ways to automatically detect and highlight unnecessary re-renders. These libraries provide detailed insights into which props or state changes triggered the re-render, helping developers optimize their components more effectively.

After reviewing performance tools and debugging techniques, candidates like Jane should be well-prepared to answer questions on how to debug and enhance application performance. Here is how she responded in a recent interview: "When I encounter performance issues related to unnecessary re-renders in a React application, I follow a systematic approach.

First, I focus on identifying the problem by carefully observing the application's behavior to pinpoint the performance bottleneck. This often appears as slow UI updates or a laggy user experience. I then use React Developer Tools—specifically the Profiler tab—to get a high-level overview of component performance and quickly identify which components are re-rendering excessively.

Next, I analyze these re-renders in more detail using the Components tab in React Developer Tools. I inspect the component tree to see which components are re-rendering and examine their props and state for unexpected changes. Common causes include unintentional prop changes, state updates that do not actually change values, and unnecessary re-renders triggered by context updates.

Once I identify the root cause, I apply targeted optimizations. For functional components, I use `React.memo` to prevent re-renders when props have not changed. I also use `useMemo` and `useCallback` to memoize expensive calculations and avoid unnecessary function recreations, especially when passing functions as props.

Finally, if I am using React 19 beta or later, I leverage the React Compiler,